

# Sina Niyavarzi

SOFTWARE ENGINEER

Hamburg, Germany

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## Summary

Software Engineer with 8+ years of experience designing and developing complex backend, frontend, and hardware-connected applications. Strong experience with **Python**, **TypeScript**, **PostgreSQL**, **REST APIs**, and **microservice architectures**, including banking systems, engineering software, and IoT platforms supporting 10,000+ devices. Experienced in reviewing existing systems, contributing to architecture decisions, and implementing maintainable solutions across the full software lifecycle. Electrical Engineering background with practical experience in real-time communication and software for machine-related and technical applications.

## Skills

|                                   |                                                                                             |
|-----------------------------------|---------------------------------------------------------------------------------------------|
| <b>Languages</b>                  | <b>Python</b> , <b>TypeScript</b> , JavaScript, Go                                          |
| <b>Backend &amp; APIs</b>         | <b>FastAPI</b> , <b>Django</b> , Flask, NestJS, <b>RESTful APIs</b> , OAuth2/OIDC           |
| <b>Architecture</b>               | <b>Microservices</b> , Event-driven Systems, Distributed Systems, API Design, System Design |
| <b>Frontend</b>                   | <b>React</b> , <b>Next.js</b> , Tailwind CSS, TanStack Query, Qt/QML                        |
| <b>Databases</b>                  | <b>PostgreSQL</b> , TimescaleDB, SQLite, MySQL                                              |
| <b>Industrial &amp; Messaging</b> | <b>MQTT</b> , RabbitMQ, Real-time Communication, IoT Systems                                |
| <b>Engineering Practices</b>      | <b>Git</b> , Agile/Scrum, Code Review, Testing, CI/CD, Performance Optimization             |
| <b>Infrastructure</b>             | Docker, GitHub Actions, Linux, AWS                                                          |

## Work Experience

### Hyphorest

SOFTWARE ENGINEER

Remote, Germany

Jan 2025 - Apr 2026

- Developed responsive application interfaces with **TypeScript** and contributed to backend architecture and scalability decisions.
- Designed reusable frontend components and shared patterns, improving performance and accessibility scores by 40%.
- Worked with backend and design teams to translate product requirements into onboarding, analytics, and platform features.

### Middle East Bank

FULL-STACK DEVELOPER

Tehran, Iran

Jul 2022 - Oct 2024

- Designed and developed multilingual banking applications and backend modules used by more than 50,000 users.
- Built **Python/FastAPI microservices**, **RESTful APIs**, and **PostgreSQL** integrations for secure authentication and data access.
- Contributed to frontend and backend design decisions, reducing page load times by over 60% and improving API performance through caching and rendering optimizations.

### Atashyar

SOFTWARE DEVELOPER

Tehran, Iran

Nov 2020 - Jun 2022

- Developed **Python** engineering software for mathematical modelling and analysis of complex physical systems.
- Designed application interfaces, processing workflows, and local data storage, reducing manual analysis time by 30%.
- Worked closely with technical domain requirements to convert engineering concepts into maintainable software modules.

### XROBOX

BACKEND DEVELOPER

Tehran, Iran

Jan 2017 - Oct 2020

- Designed and developed **REST APIs** and backend services for IoT platforms supporting more than 10,000 connected devices.
- Implemented high-volume, real-time communication between devices and backend systems using **MQTT** and event-driven processing.
- Designed time-series data storage and distributed processing workflows, improving the scalability and reliability of hardware-connected applications.

## Projects

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### Spectrometer Simulator [snnyvrz/spectrometer](#)

2026

[Python](#) • [FastAPI](#) • [Next.js](#) • [WebSockets](#)

- Developed a real-time spectrometer simulator using **FastAPI**, asynchronous Python, WebSockets, and **Next.js**.
- Implemented synchronized backend simulation and live frontend visualization for interactive measurement workflows.

### ShelfShare [snnyvrz/shelfshare-infra](#)

2025

[Go](#) • [Express](#) • [Kubernetes](#) • [Helm](#) • [FluxCD](#) • [PostgreSQL](#)

- Built a cloud-native microservice platform for book management with independent authentication and catalog services.
- Designed production-style infrastructure using **Kubernetes**, **Helm**, **FluxCD**, Docker, GitHub Actions, and GitOps workflows.

## Education

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### SRH University

*Hamburg, Germany*

M.A. DIGITAL TRANSFORMATION MANAGEMENT

- Admission ranking: 119th among approximately 10,000 applicants.
- Achieved grade 1.0 (highest grade) in Digital Business Analytics.

### Sharif University of Technology

*Tehran, Iran*

B.SC. IN ELECTRICAL ENGINEERING

- National university entrance exam rank: 84th among approximately 200,000 candidates.
- Achieved one of the highest grades in the C++ programming course.

## Professional Training

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### AWS Cloud Technical Essentials

2024

AMAZON WEB SERVICES

- Learned the fundamentals of **cloud computing** and **AWS architecture**, including **EC2**, **S3**, **RDS**, **IAM**, and **VPC**.
- Gained an understanding of designing **secure**, **scalable**, and **highly available** cloud applications following AWS best practices.

### Easy to Advanced Data Structures

2019

UDEMY

- Studied fundamental and advanced **data structures** including arrays, linked lists, stacks, queues, trees, heaps, graphs, and hash tables.
- Applied **algorithm design**, **time and space complexity analysis**, searching, sorting, and problem-solving techniques.

### Mathematical Thinking in Computer Science

2021

COURSERA

- Developed strong foundations in **discrete mathematics**, **logic**, **proof techniques**, **recursion**, **graph theory**, and **combinatorics**.
- Strengthened analytical reasoning and mathematical skills used in **software engineering**, **algorithms**, and **computer science**.

### Java/Android Development Bootcamp

2019

MAKTAB SHARIF

- Built Android applications using **Java**, applying object-oriented programming, Android components, and standard mobile UI patterns.
- Implemented practical features such as user input, navigation, local data storage, and API integration through hands-on projects.

## Languages

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- English ([C1](#), [IELTS 7.5](#))
- German ([B2](#), [Currently Learning](#))
- Persian ([Native](#))